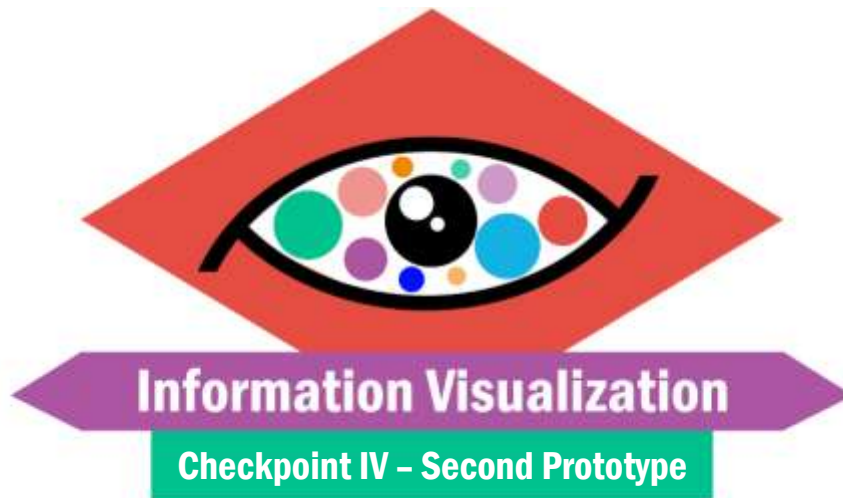




Objective

At the end of this checkpoint, you have a functional version of your visualization mainly consisting of creating the layout and implementing **at least two of the idioms** you have selected on Checkpoint II.

Requirements

Continue the work you started in the previous checkpoint. You are required to implement **at least two idioms**. This includes all the integration and interaction that makes them work together as part of a single, coherent dashboard.

All the requirements of this checkpoint are the same as the previous one, but now you need to have at least two visual idioms.

Again, but important, **you are only allowed to use D3.js**. No other frameworks are permitted. This means do not use React, Angular, Node.js, or any similar tools. If you do, your checkpoint will receive a **grade of 0**. You may use Python (and Python alone) for the backend, if necessary, but your frontend must be built using only plain JavaScript, HTML, and CSS. No TypeScript, no transpiling, no SASS, etc. Also keep in mind your prototype must be stand-alone so if you use any JS libraries, etc. make sure you have a local copy. Rule-of-Thumb: if it works without an internet connection, you're probably fine.

Deliverables

Create a **video with a maximum of 3 minutes** and submit it online, until two days before your class (ex: classes on Friday must submit until Wednesday at the end of day). Use either the h264 or h265 codec, and a modern container (mp4, mkv, etc.). Weird stuff we cannot play with the default installs of Windows/MacOS/Debian will not be graded. Name your video '**CPIV-<group number>.<extension>**' (the extension will depend on the container format you choose).

Submit it online, **inside a zip file including both the video and the code, in Moodle**.

The **video** should describe and illustrate **only deltas from the previous checkpoint regarding**:

- The **code architecture** you have created to load and (pre)process the data, to make the charts (and their interactivity), and to handle the integration.
- The **layout of your visualization**. Include at least one shot of the complete interface (yes, even the idioms you have not implemented, in which case use screenshots from Checkpoint II).
- The **data (pre)processing** you must do before the data is ready to be loaded to the charts. We are not talking about the data processing from CPI but all the remaining process you have to

do for each visual idiom.

- The idioms you have implemented (**chart visualization**) and their functionality (**chart interactivity**). You must describe them and explain the interactivity supported by those idioms.
- The **integration** techniques used and why adding more charts is easy. How are the views linked? How does that mechanism work/will it work even when you have more idioms to link?

If there are any changes from Checkpoints I and II, and III describe them and justify why you made those changes using the visualization principles learned in class and **VisDeck card spreads**. You will need to defend those spreads to validate the changes to your design. So, it is not enough to just choose an idiom you think easy or cool. The spread must match it to your questions and dataset. **Changes not justified will be penalized!**

Your prototype must **include the dataset**. You should deliver in a plug-and-play fashion: we run a local server, and the prototype is ready to be tested.

Name your deliverable as 'CPIV-<group number>.zip'. For instance, 'CPIV-02.zip'. The zip should contain the report and your code. Make sure your PDF file follows the same name nomenclature.

Penalties

- Video over 3 min long: **1 grade point penalty per extra 10sec.**
- Video uploaded after the deadline: **0.5 grade points penalty per hour of delay.**
- Incorrect file type or name: **0.5 grade points penalty per file or name.**
- Video file and ZIP names are different: **0.5 grade points penalty.**
- Incorrect file type or name: **0.5 grade points penalty per file or name.**
- Zip with video and code uploaded after the deadline: **0.5 grade points per hour of delay.**
- No prototype: **10 grade points penalty.**

Tasks to perform during the lab

Show your working prototype to the professor for **feedback**.

The grade **will be made known one week later** (see below). If you are not receiving feedback, then you must be peer assessing your colleagues' submissions.

Grading

Your work will be graded according to the following parameters:

- **Video (5%):** within allotted time, easy to understand, follows suggested structure
- **Prototype architecture (5%):** describe and illustrate **deltas** regarding the code architecture to produce the prototype.
- **Dashboard layout (5%):** present and depict **deltas** regarding the work done so far according to your Checkpoint II.
- **Additional data processing (5%):** describe **deltas** regarding your additional data processing techniques to achieve the result, e.g., `groupby` or `count` in runtime. If you do not have additional data processing, explain why. Also describe changes to the visual idioms and justify them with VisDeck.
- **Chart visualization (25%):** show **deltas** regarding the actual visualization plotting real data, with a polished look-and-feel.
- **Chart interaction (20%):** present **deltas** regarding the code modules used to achieve chart interaction.
- **Chart integration (30%):** describe **deltas** regarding the integration architecture between charts.
- **Peer assessment (5%):** grade your peers' submissions in the class.

An important note on grading for this and all other Checkpoints: always *justify* your choices, based on the basic principles you have learned in this course (adequacy of channels to data types, human perception, etc.). Don't just describe what it is, but especially *why* it is as it is.